

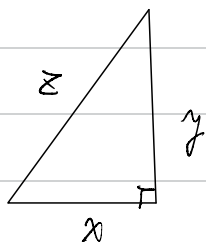
which are distinct, are pairwise distinct. So there are exactly 8 prime factors of p , given by π , $\bar{\pi}$ and their associates.

As a consequence, there are exactly 8 ways of expressing p as a sum of two squares:

$$p = \underbrace{(\pm x)^2 + (\pm y)^2}_{4 \text{ choices}} = \underbrace{(\pm y)^2 + (\pm x)^2}_{4 \text{ choices}}$$



Pythagorean triples.



$$x^2 + y^2 = z^2, \quad x, y, z \in \mathbb{Z}.$$

Eg. $(x, y, z) = (3, 4, 5), (5, 12, 13).$